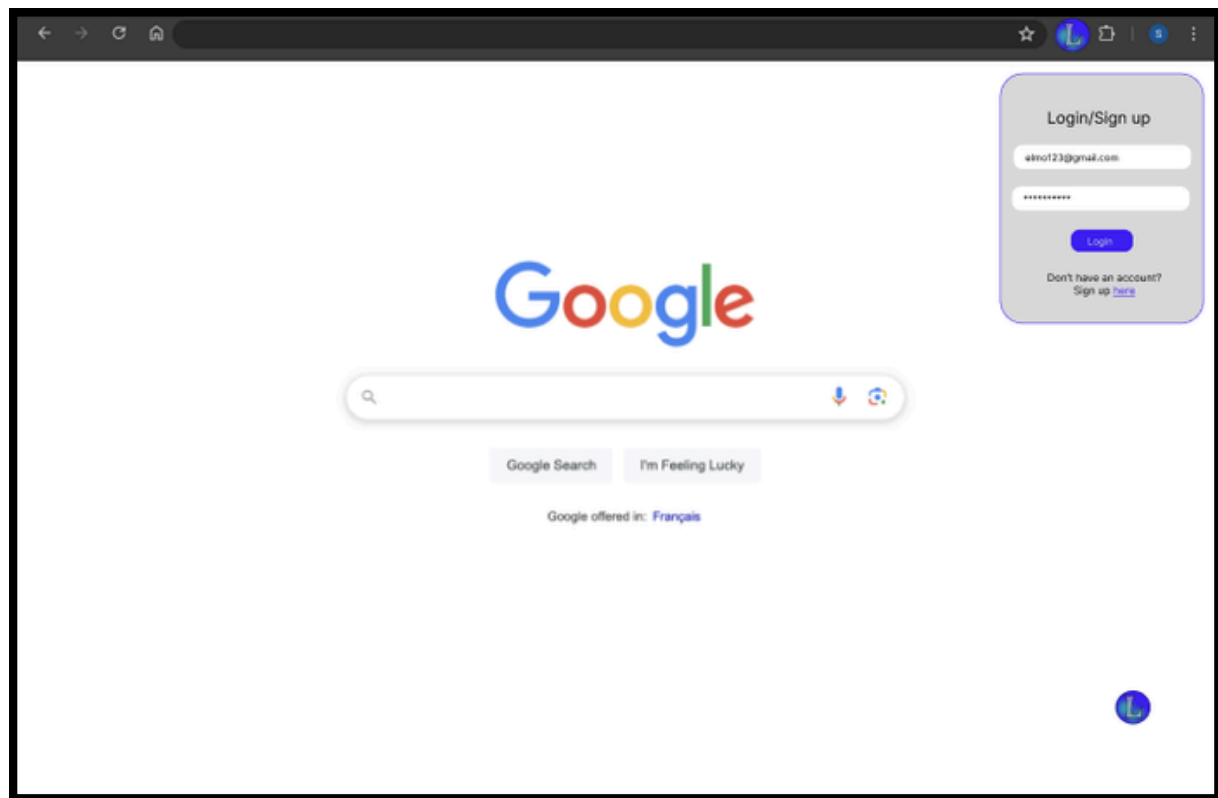
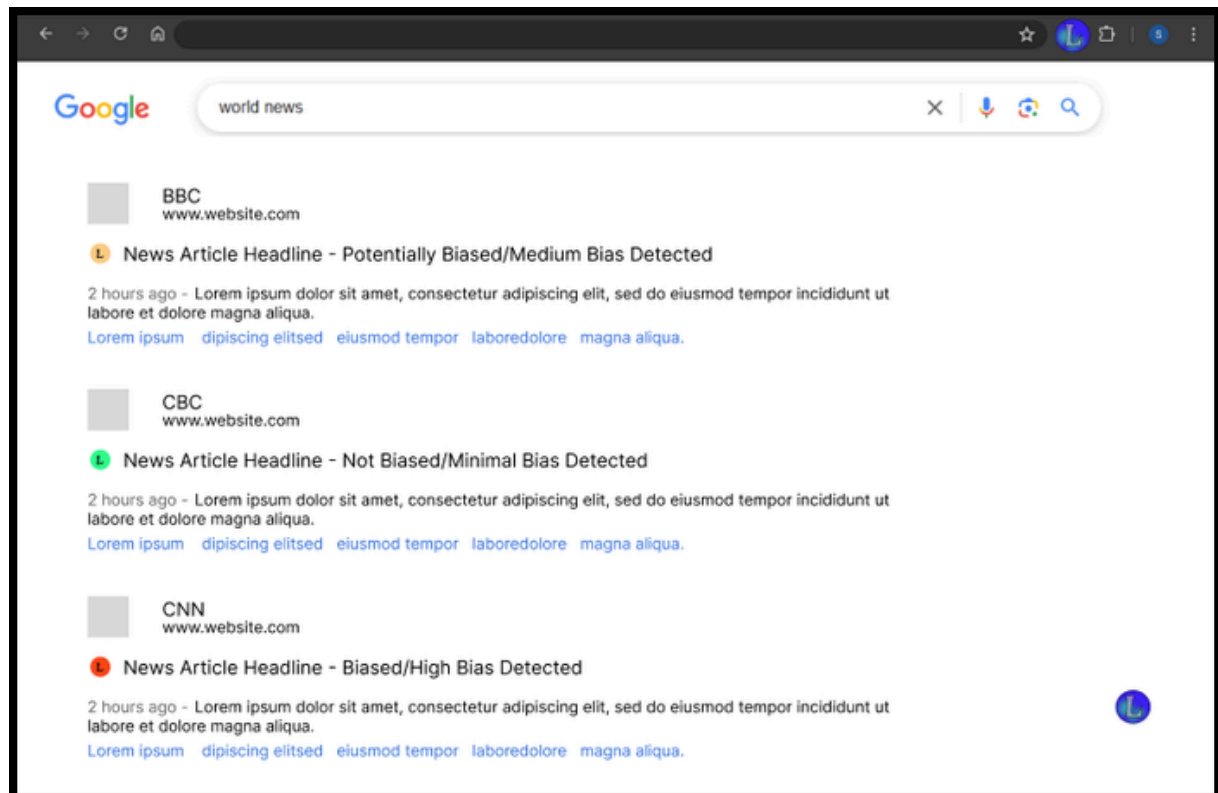


## Low Fidelity Prototypes

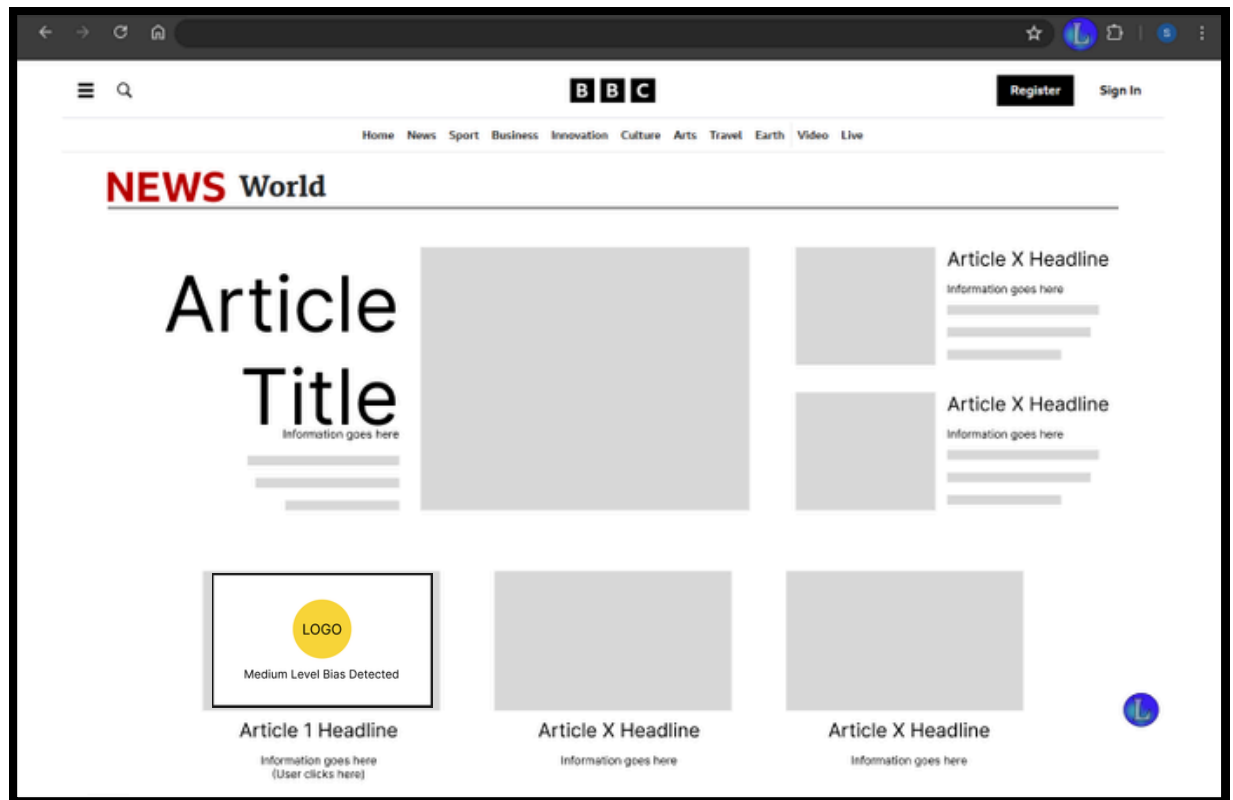
Login and Sign up  
Screen Appears  
as a drop down  
menu below  
pinned extension



Bias indicators  
appear beside  
article headlines  
giving a brief  
insight



## Low Fidelity Prototypes



Article  
Thumbnails  
appear with Bias  
ratings when  
hovering



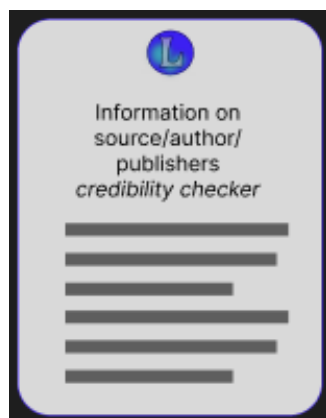
Clicking extension  
provides users  
with an Author/  
Publisher  
background

## Low Fidelity Prototypes

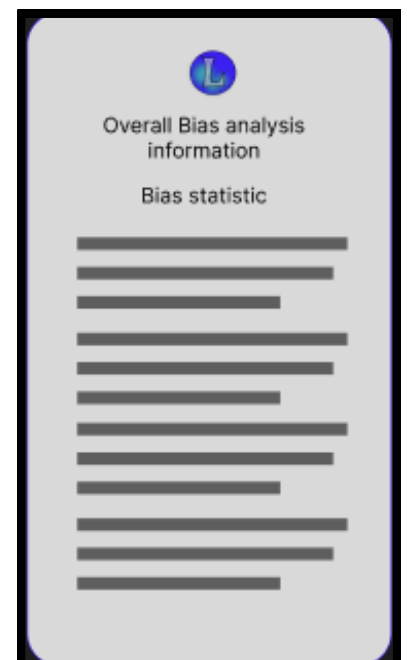
Users can highlight text for a specific bias analysis on that text passage



Source, Author, and publisher background credibility check



Bias Analysis pop-up delivered when highlighting or when first browsing



## User Personas

### DYLAN

#### PROFILE

Gender : Male  
Age : 21  
Education : Political Science Major  
Occupation : University Student  
Location : Toronto



#### PAIN POINTS

- Struggles with finding balanced news sources
- Feels overwhelmed by misinformation on social media

#### GOALS AND EXPECTATIONS

- Be well informed and prepared for debates and discussions in class
- Seeks credible sources that aren't overly biased
- Easily find credible sources without searching through tons of biased articles first
- Find an easy, time-efficient way to compare perspectives without doing manual research

#### BEHAVIOUR

- Actively follows and engages with political news on Twitter, Reddit and Youtube
- Subscribes to multiple news sources but struggles with knowing which are biased
- Regularly uses fact-checking software like Snopes and FactCheck.org
- Participates in university discussions, debates and social activism groups
- Reads comment sections to gauge different perspectives, especially of differing opinions

#### OPPORTUNITY / HOW PRODUCT OR SERVICE HELPS

- Highlights level of bias in articles and other media sources to quickly assess credibility
- Suggests alternative sources with different perspectives that save time in research process
- Provides quick summary of bias trends without overwhelming user with information
- Gives user control over depth of analysis, allowing him to highlight sections for detailed breakdowns of the text

### RACHEL

#### PROFILE

Gender : Female  
Age : 25  
Education : Bachelor's Degree  
Occupation : Nurse  
Location : Waterloo



#### PAIN POINTS

- Encounters medical misinformation on social media
- Can't tell the difference between fake AI generated news

#### GOALS AND EXPECTATIONS

- Stay up to date with global healthcare news
- Improving digital literacy to understand which media is AI generated
- Avoid AI-generated / misleading articles, especially during health outbreaks

#### BEHAVIOUR

- Subscribes to healthcare journals or newsletters to stay updated on healthcare trends
- Engages in discussions about AI in healthcare with peers
- Educates patients on misleading information on the internet
- Scrolls through social media during long shift breaks and views lots of medical related posts
- Follows healthcare influencers to stay engaged about relatable work

#### OPPORTUNITY / HOW PRODUCT OR SERVICE HELPS

- Highlights biased, AI-generated, or misleading information in health news to help her make informed decisions
- Saves user time by providing a quick credibility check
- Gives user a different perspective to political news that impacts the healthcare industry

## User Personas

### ALEX

#### PROFILE

Gender : Non Binary  
Age : 26  
Education : Bachelor's Degree  
Occupation : Elementary Science Teacher  
Location : Sudbury



#### PAIN POINTS

- Educating students on the disinformation on social media
- Having biased / sources for writing papers
- Finding un-biased sources for reliable information

#### GOALS AND EXPECTATIONS

- Teach students how to identify reliable information
- Provide honest and transparent classroom content
- Avoid spreading misinformation to students
- Stay informed on science education trends

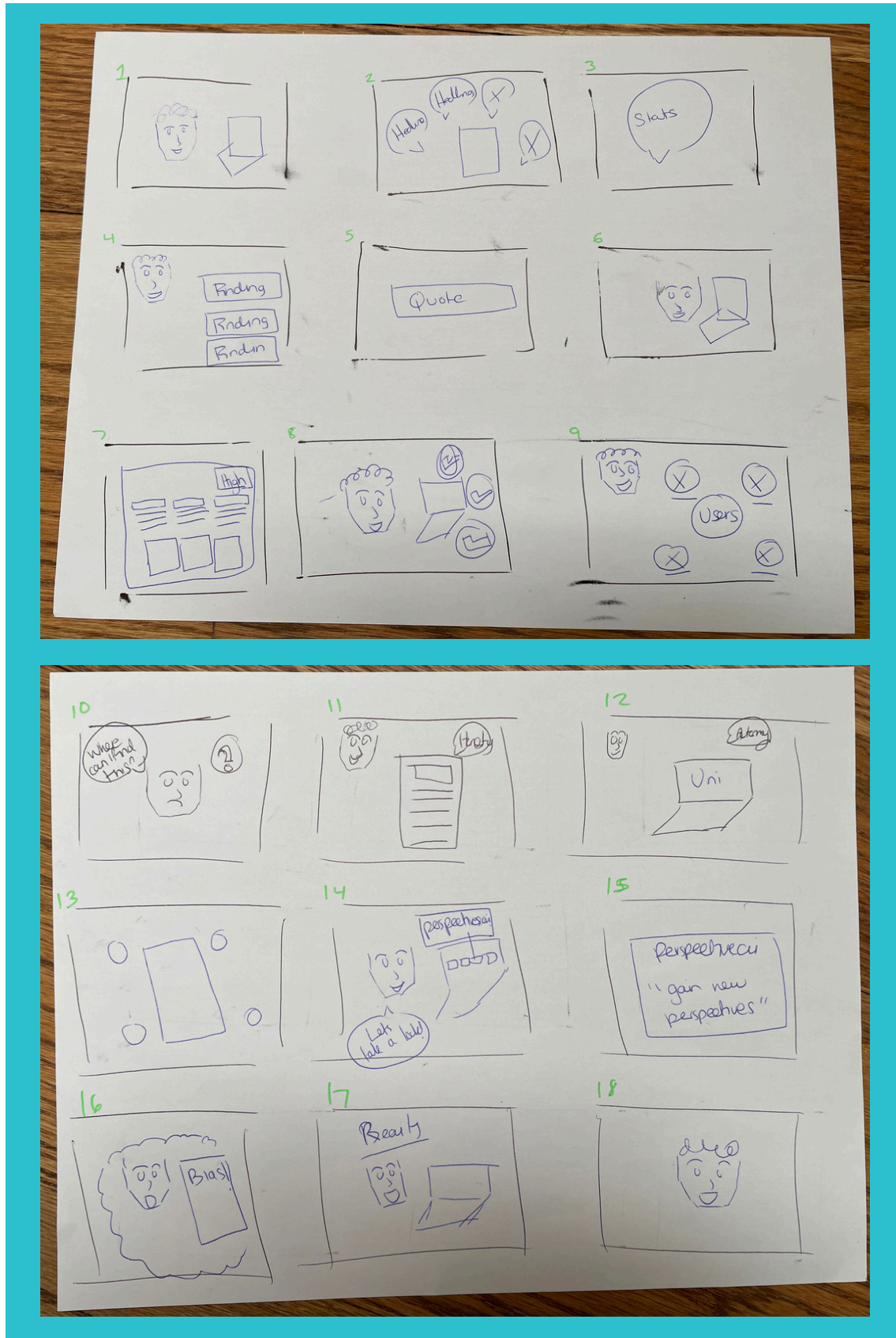
#### BEHAVIOUR

- Reads science and education related news to keep up with changing trends
- Follows and engages with other teachers on social media, like Twitter threads
- Uses fact checkers like Snopes before incorporating information to lesson plans
- Watches televised news in the evenings and receives article news from Snopes

#### OPPORTUNITY / HOW PRODUCT OR SERVICE HELPS

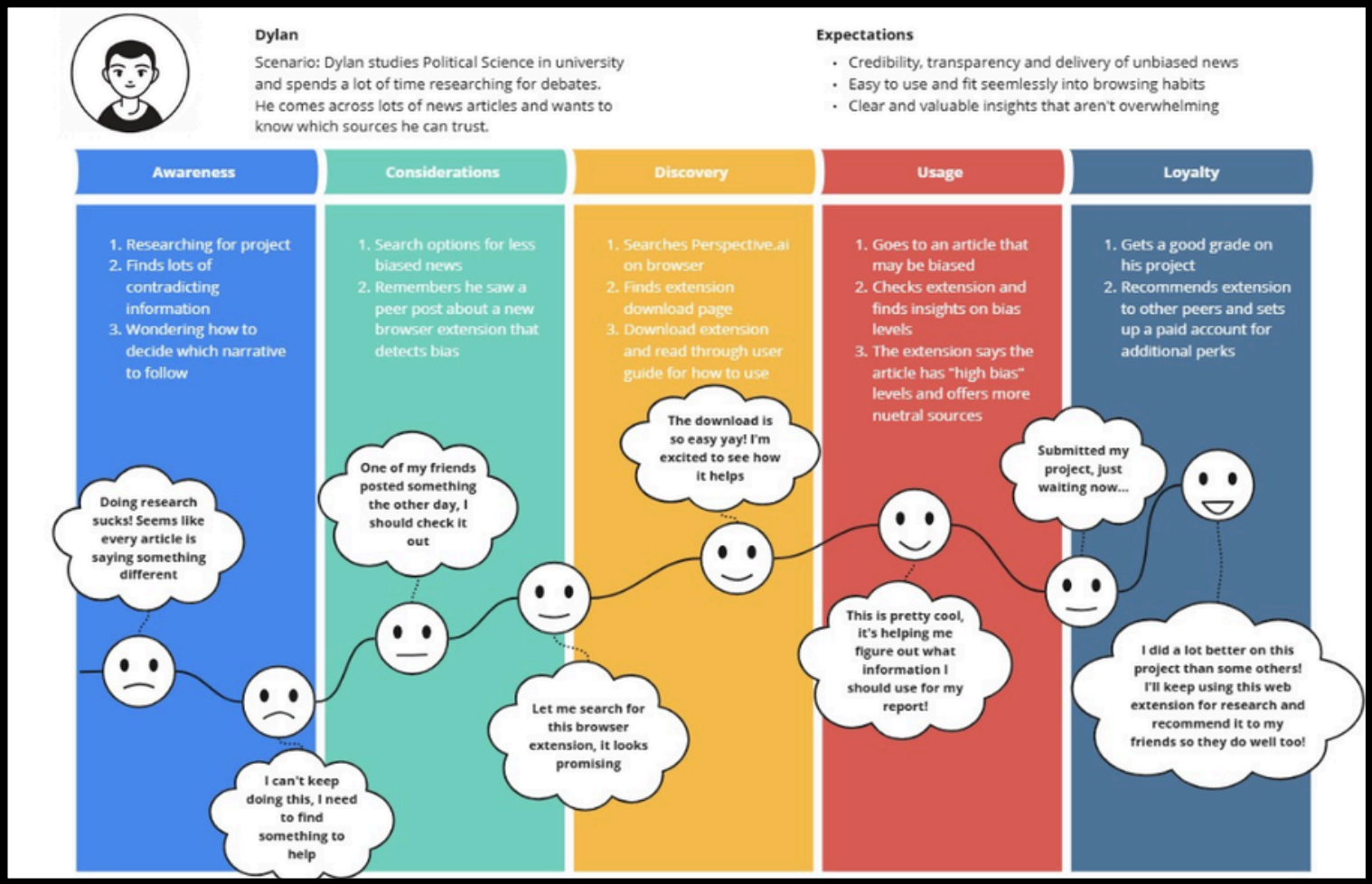
- Teach students how to use service to get perspectives on information
- Identifies bias in science related news to make researching for lesson plans easier
- Suggests diverse sources to provide many perspectives easily accessible in one place

## MVP VIDEO Storyboard

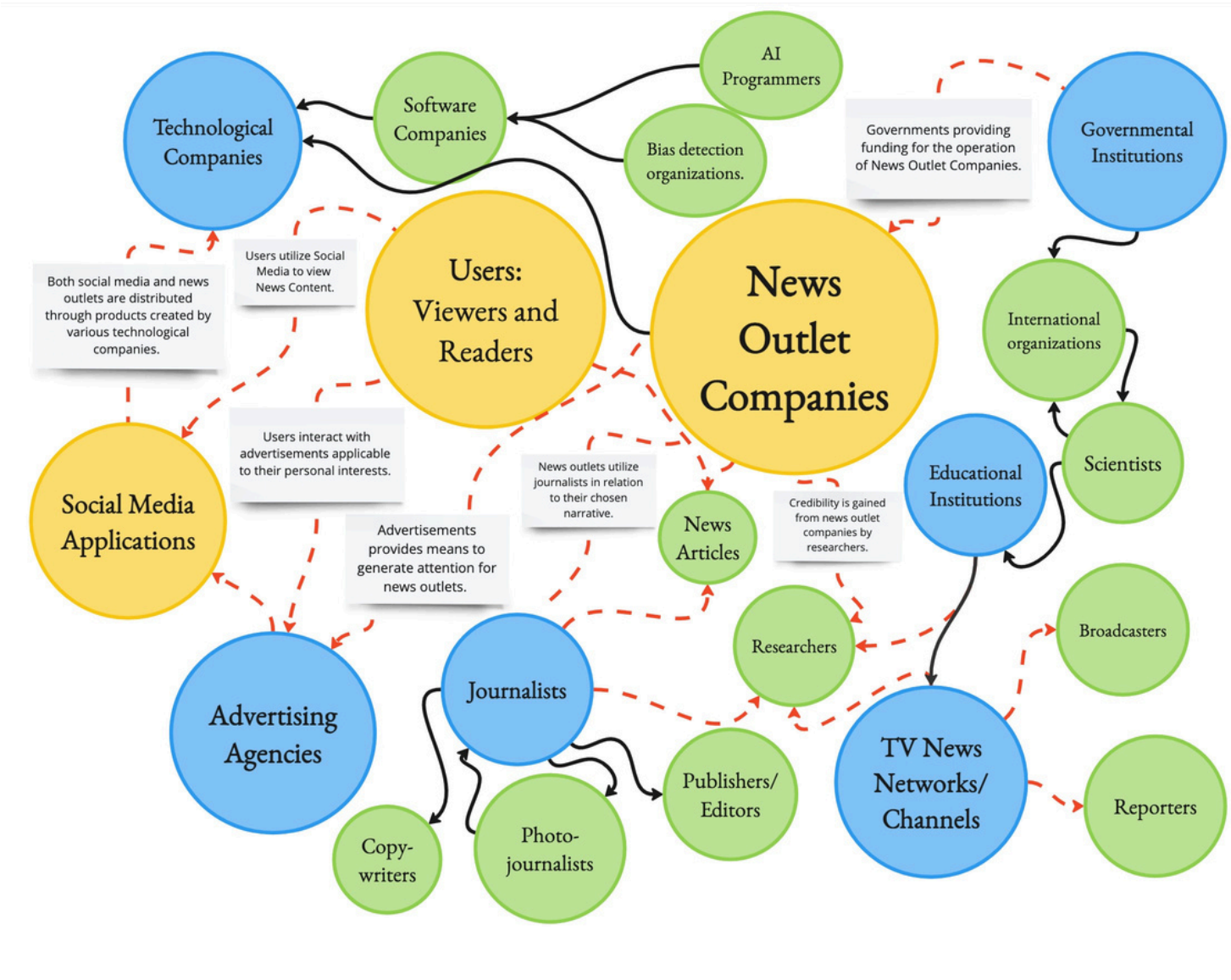


# Appendix

## User Journey Map



## Stakeholder Map



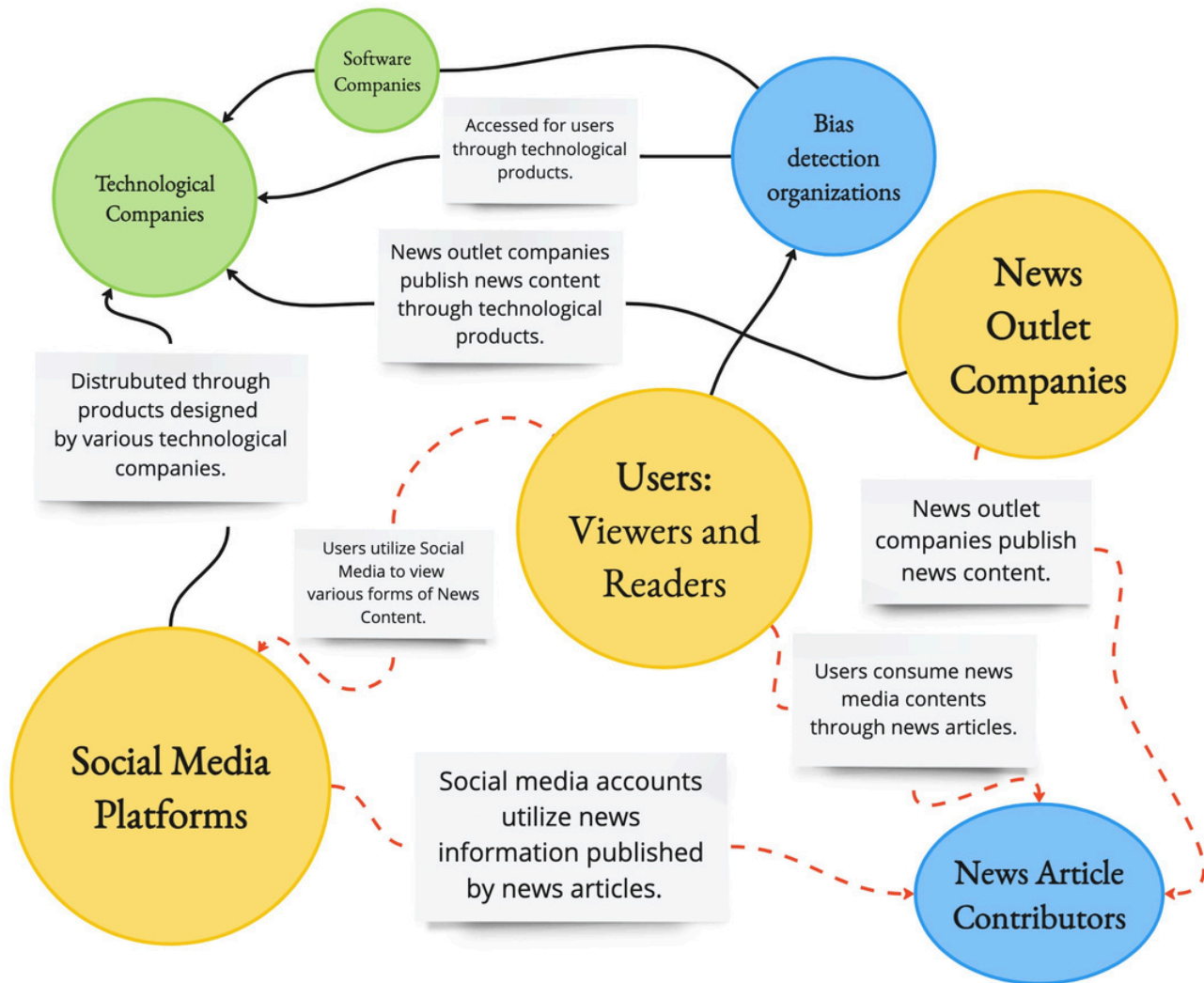
As displayed within the system map, relationships between stakeholders have been identified to better understand bias in news media, and the interactions between various stakeholders.

### System Map Legend:

- Primary Stakeholders
- Secondary Stakeholders
- Background Stakeholders
- Unbiased relationship
- - - - -> Bias at play

The **direction of the arrow** indicates the dependency of a stakeholder on the other.

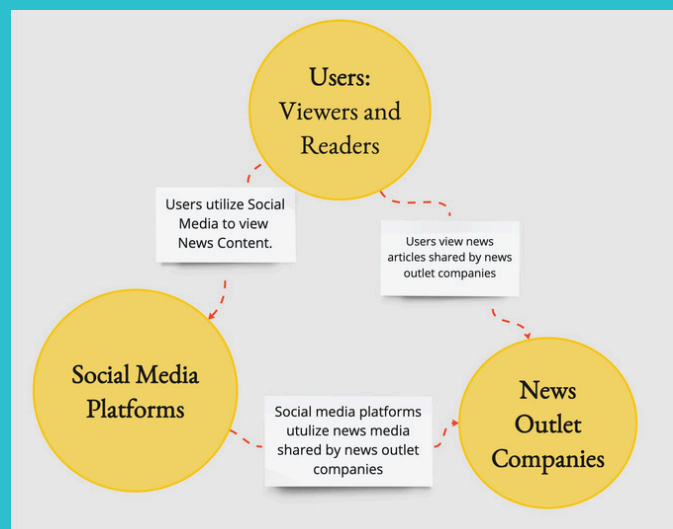
## Key Primary Stakeholders



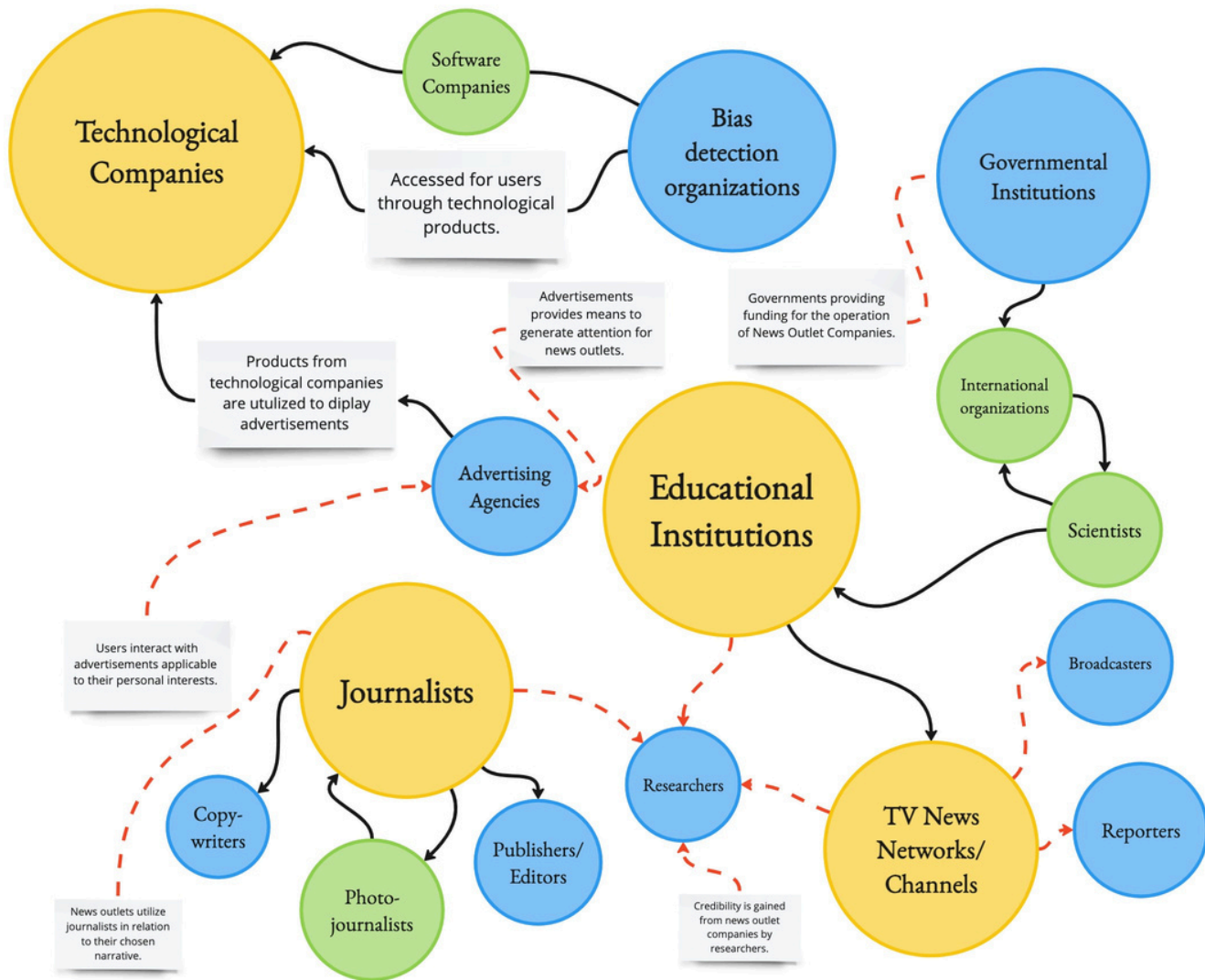
### System Map Legend:

- Primary Stakeholders
- Secondary Stakeholders
- Background Stakeholders
- Unbiased relationship
- - -> Bias at play

The **direction of the arrow** indicates the dependency of a stakeholder on the other.



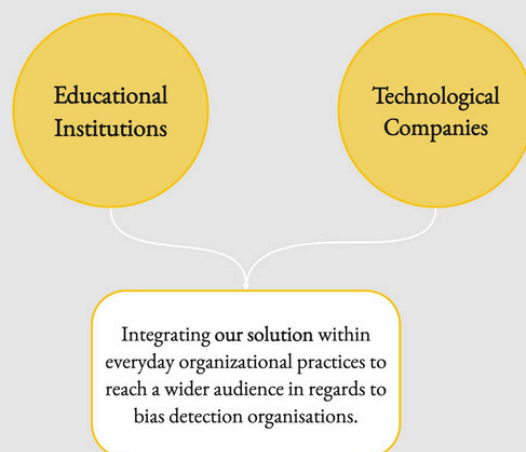
## Key Potential B2B Stakeholders



### System Map Legend:

- Primary Stakeholders
- Secondary Stakeholders
- Background Stakeholders
- Unbiased relationship
- - → Bias at play

The **direction of the arrow** indicates the dependency of a stakeholder on the other.





# Appendix

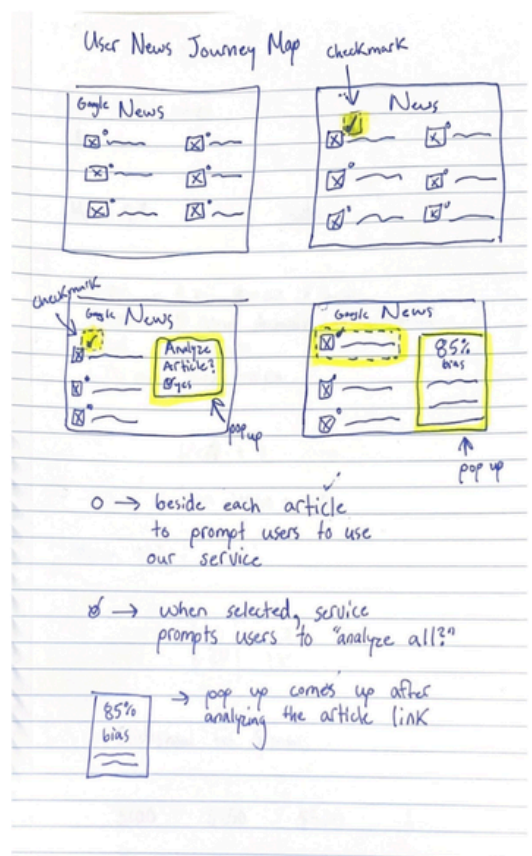


Figure 1

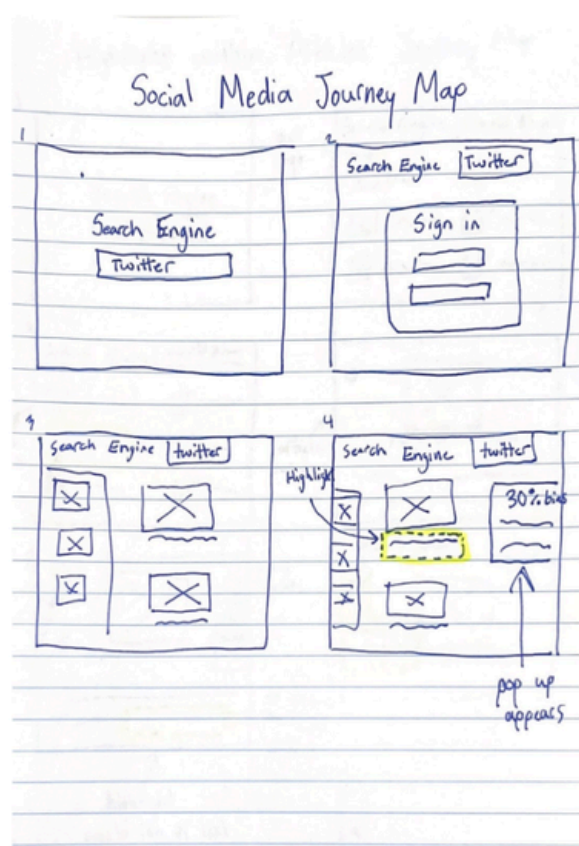
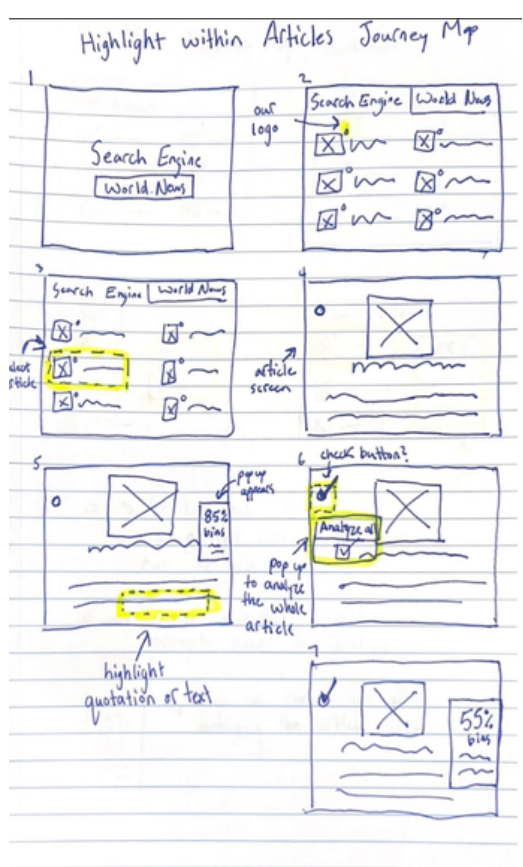


Figure 2



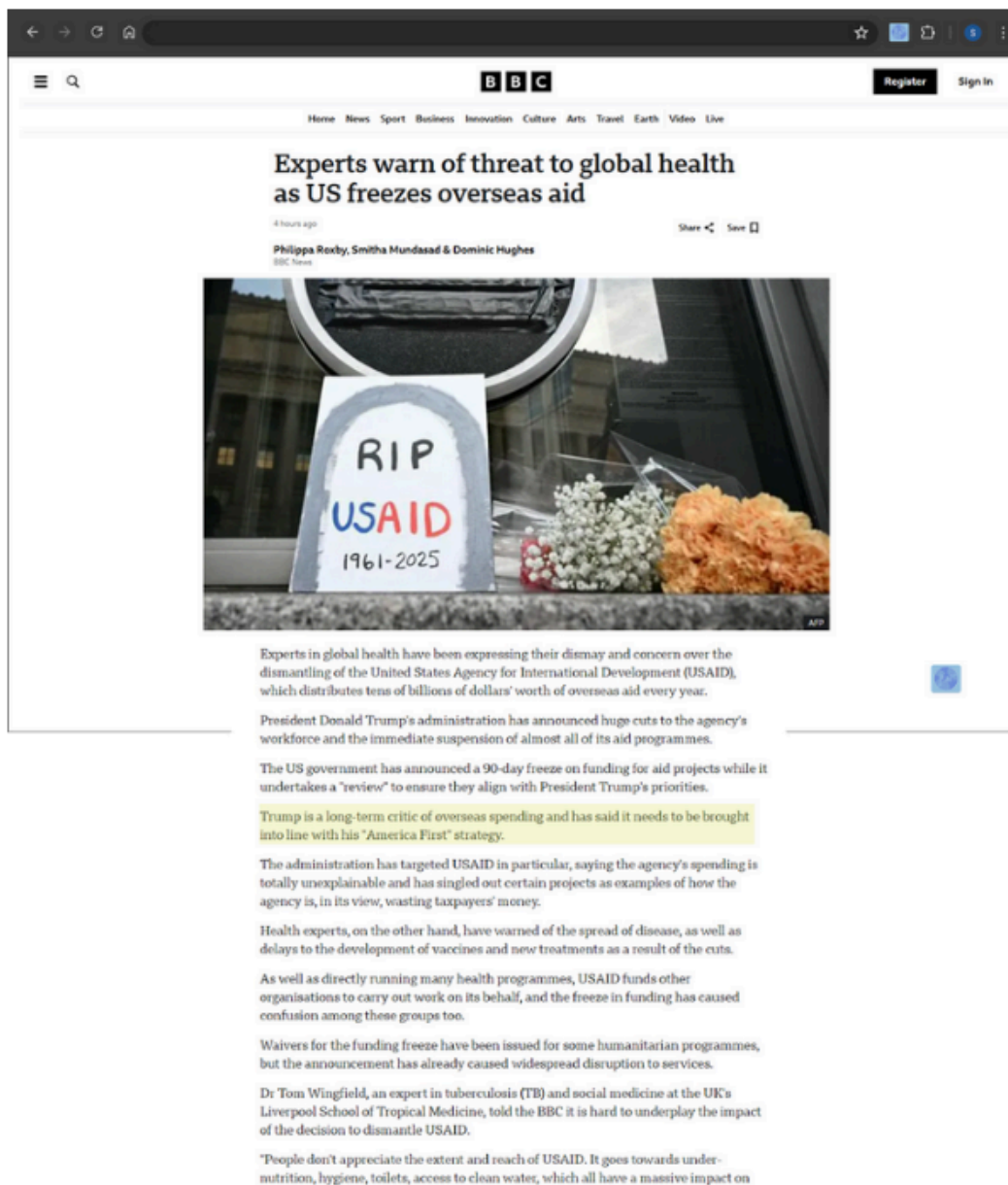


Figure 4

# Appendix

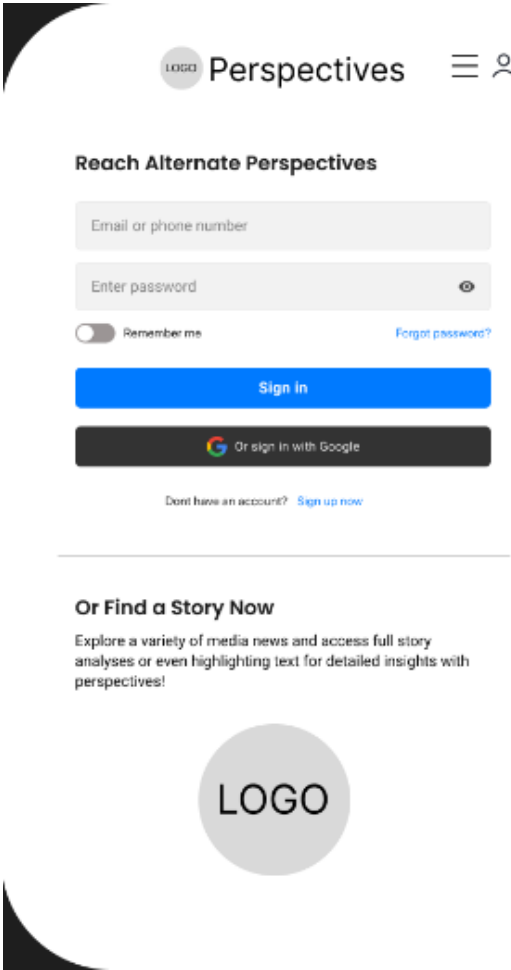


Figure 5

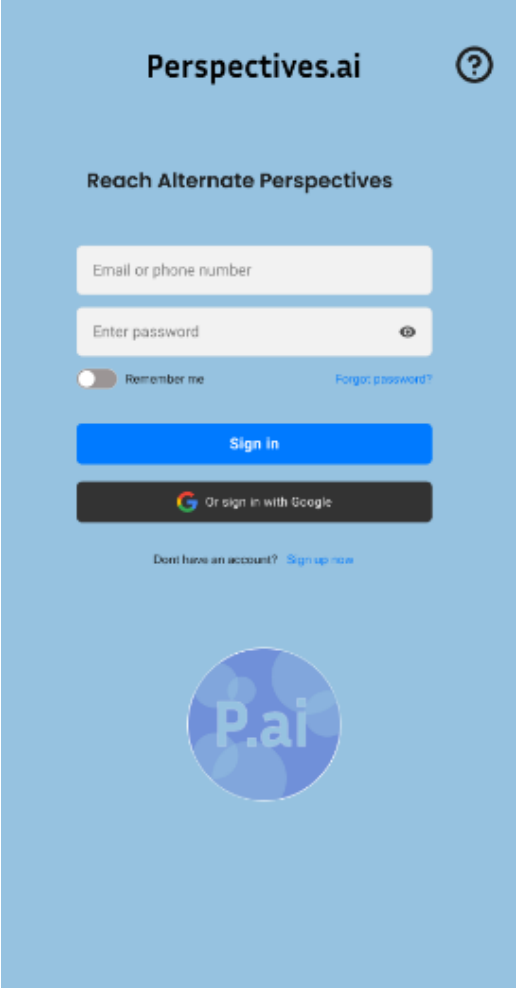


Figure 6

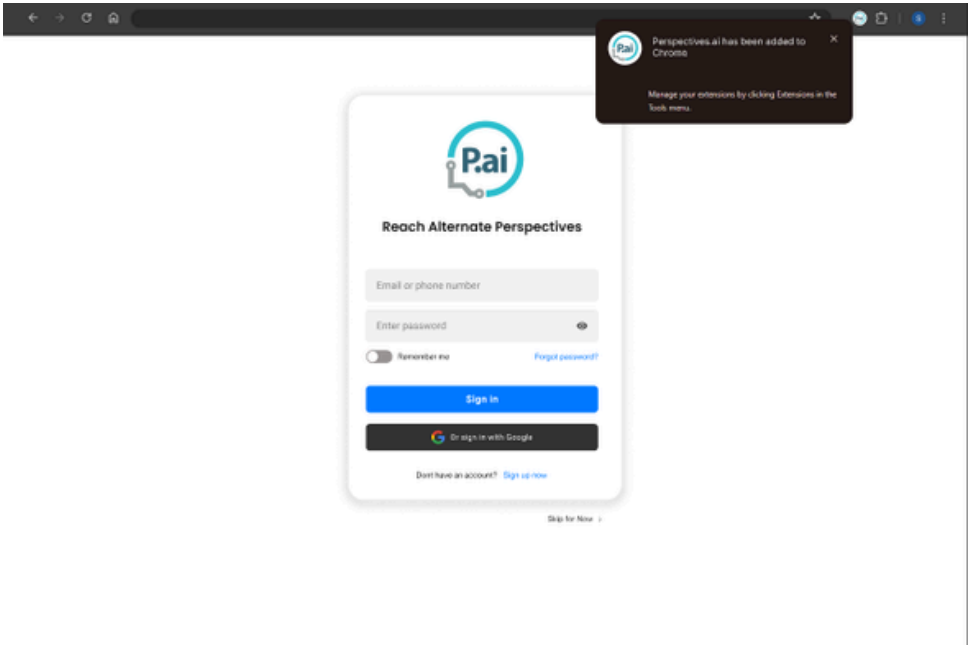


Figure 7

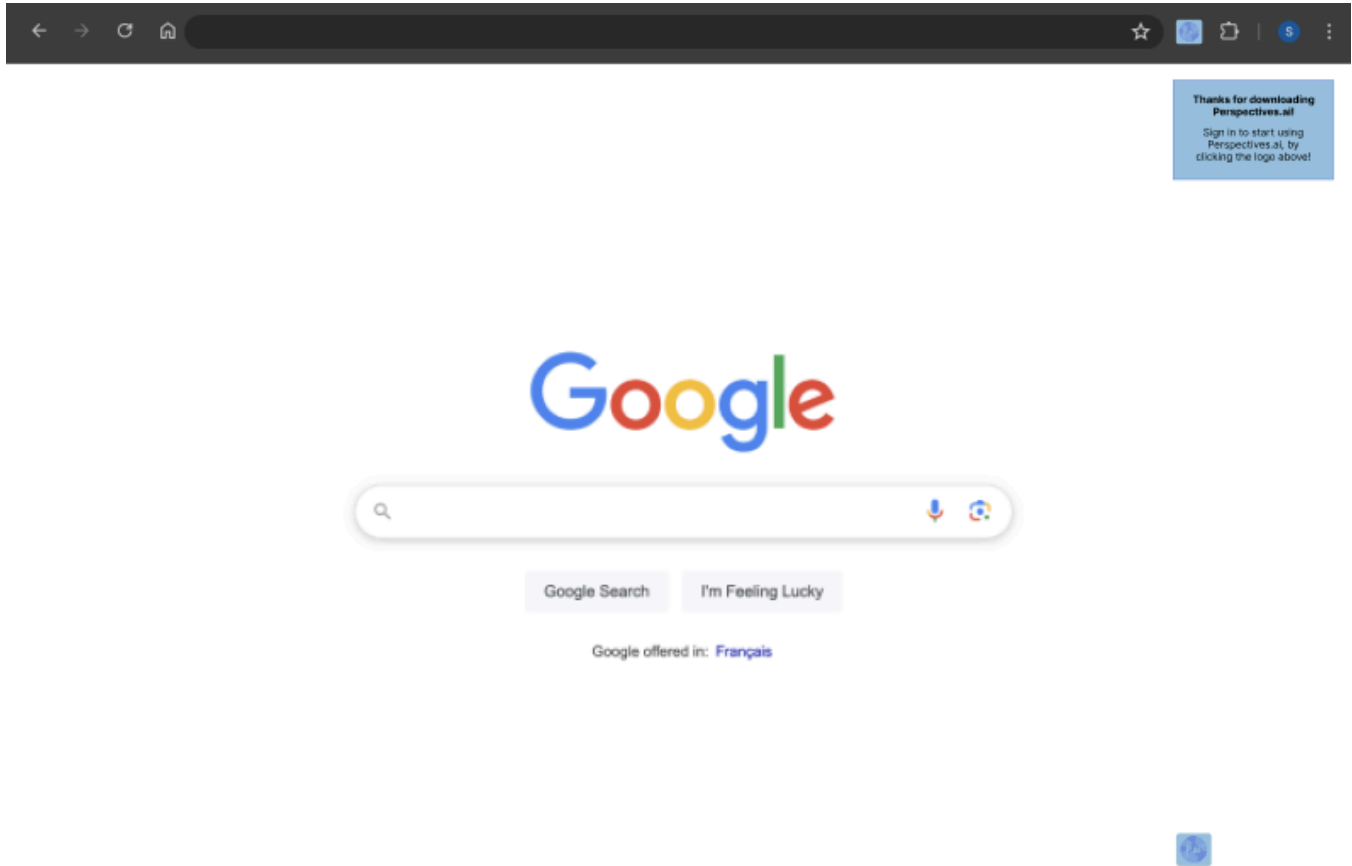


Figure 8

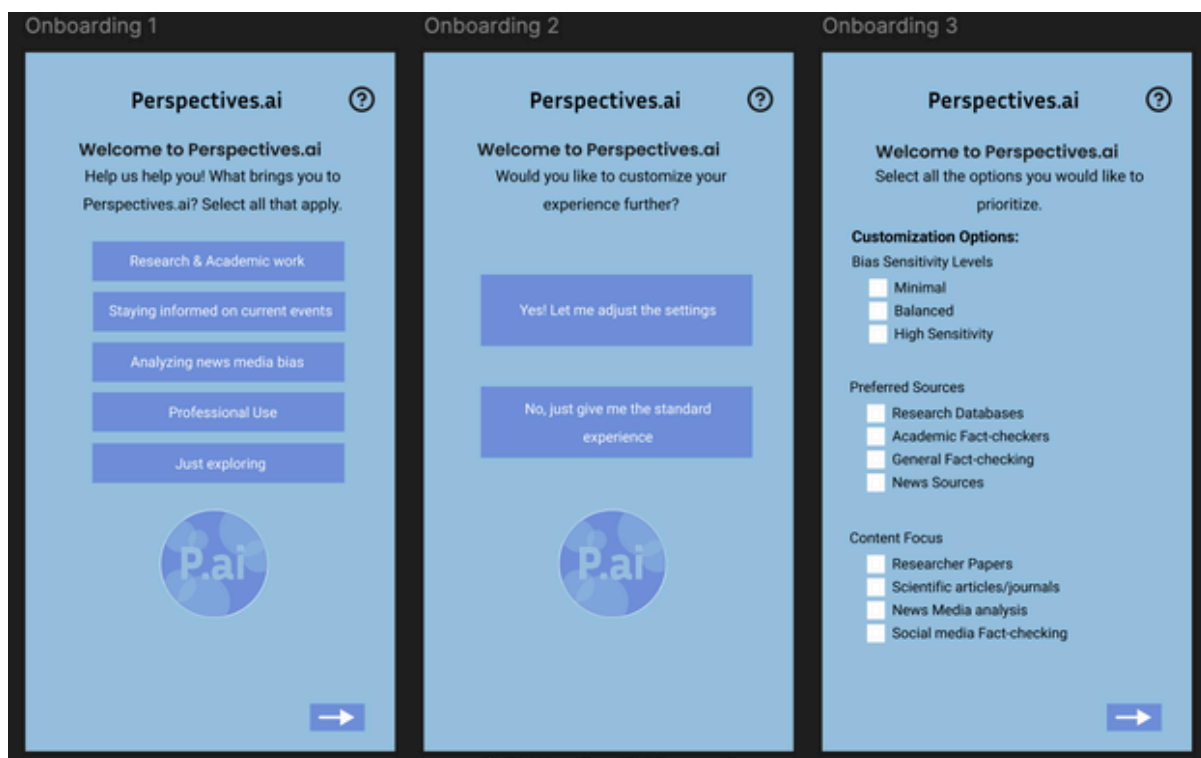


Figure 9

# Appendix



Figure 10



Figure 11



Figure 12

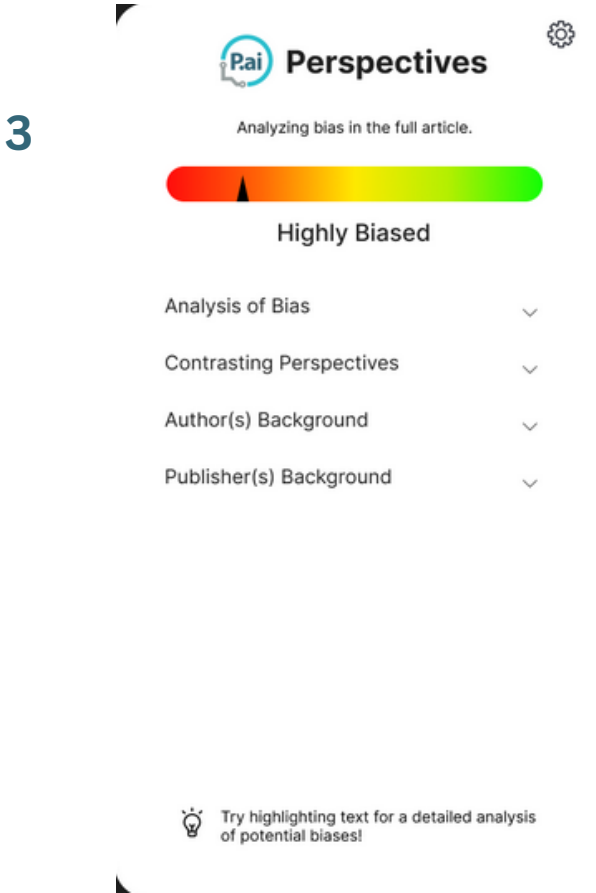
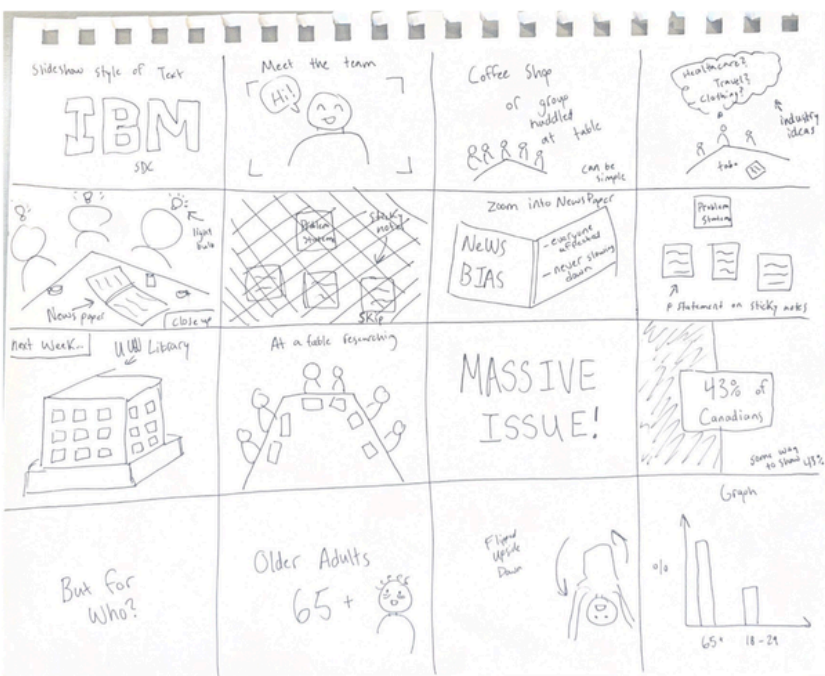


Figure 13

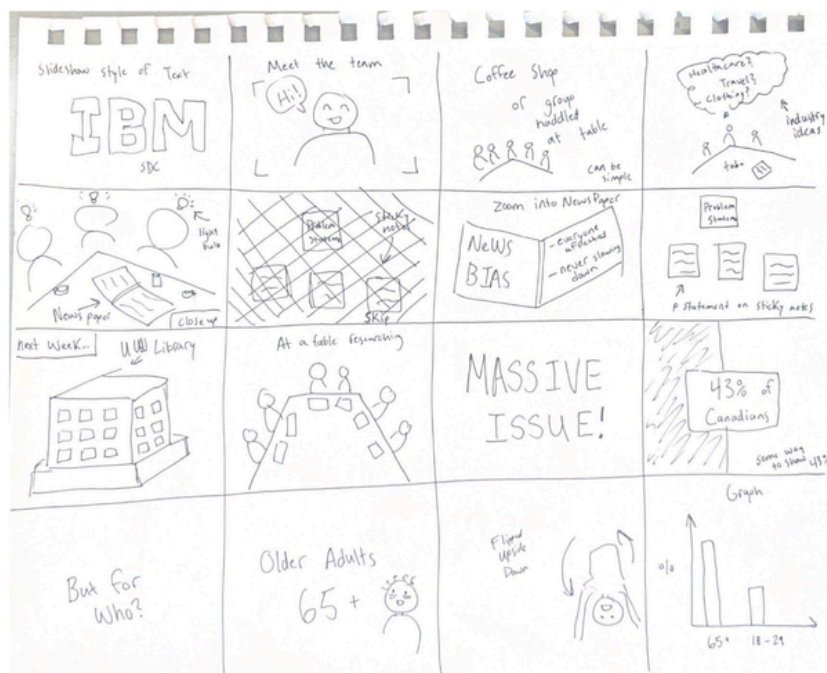
# Appendix

## Final Video Storyboards



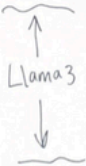
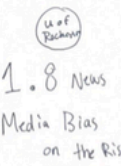
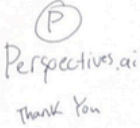
# Appendix

## Final Video Storyboards



# Appendix

## Final Video Storyboards

|                                            |                                                                                   |                                      |                                                                                    |                                                                                                    |
|--------------------------------------------|-----------------------------------------------------------------------------------|--------------------------------------|------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| 2 Primary Functions                        | 1: Highlight                                                                      | 2: Article Analysis                  |  | Not influence what you read                                                                        |
| AI? → Llama3                               |  | BIAS in AI?                          |  | NEED SOLUTIONS  |
| Improve Digital M Litter<br>Bias Awareness |  | The Goal of Perspective.ai is not... |  |                                                                                                    |
|                                            |                                                                                   |                                      |                                                                                    |                                                                                                    |

## User testing for prototype:

### Usability testing instructions:

We are aiming to better understand our web-extensions usability, target demographics behaviour, target demographic preferences, and needs. Moreover, from the usability test we hope to identify any issues surrounding our web-extension and how we can improve it.

### Usability test setup:

- We will perform this test through a remote process, and will be an moderated test.
- Please make sure to follow the testing process for consistency throughout our testing.
- Our participants are aimed at students.
- We are hoping to better understand how our users are interacting with our web-extension, and additionally gain feedback.
- Before the start of the usability test, participants will be asked for their permission to record the session and obtain the ability to leave at any-time during the test.

### Usability test preparations:

- Take the time to go through our prototype to make sure it is functioning smoothly.
- Whilst conducting the usability test please make sure to take general notes of any observations, behaviours, thoughts, or feedback during the usability testing.
- Make sure all technical aspects of the session are working prior to the usability test, making sure the mic is clear for both the moderator and participant.

### Usability test procedure:

1. The test will begin by reading the "Test script" to the participant.
  - a. The participant will be recorded after the test script is completed.
2. The prototype link will then be sent to the participant. [Link](#) (Please make sure to tell the participant to fill screen so that they can view the prototype properly)
  - a. Let the participant know that the prototype simply functions by clicking through the screens.
  - b. Please inform the user of how the prototype works if they are unfamiliar with prototype interactions.
  - c. Ask the participant to **share their screen** once they open the prototype to better understand their interactions or provide assistance if help is needed.
3. Remind the participant to act natural throughout the test.
  - a. Encourage the participants to navigate through the prototype how they normally would when researching for educational purposes.
  - b. Encourage the participants to think out loud, and verbalize any thoughts they may have, and how certain aspects of the prototype makes them feel specifically in educational settings.
4. The "Test plan" will then be read to the participant.
5. Begin the test and subtly guide the user throughout the process.
  - a. Please make sure the recording function has been turned on.
  - b. Please make sure the participant shares their screen after the prototype is successfully opened.
  - c. The participant can end the screen sharing once they have navigated through the entire prototype.
6. Have the user complete the SUS survey

7. After the "Test plan" the participant will be asked the "Follow-up questions."
8. After follow-up questions are completed thank the participant and end the recording.

## **Test script:**

Hello!

Thank you for joining me today, and taking the time to participate in our usability test for our prototype. So as you may have been informed this session is being conducted for a university project, in which we are hoping to explore the various ways to tackle bias. In collaboration with my group and I, you will be helping us gain further insights about our project, where we are exploring the presence of bias in the media.

For today, your main task will be to interact with our prototype being a web-extension that has been designed to aid students in detecting bias within their educational activities. Upon interacting with the prototype we would like you to imagine how you would feel as someone who may use our web-extension for certain research processes within an educational setting, and how our tool is able to fulfil this need to successfully understand forms of bias within the media. Please remember that we are testing our web-extension, and not how well you are able to navigate through the prototype as your feedback will help us to further improve aspects of the bias detection tool.

## **Test plan script:**

1. With the prototype opened, try to imagine how you would feel utilising the web-extension for research purposes.
2. Remember, there is no right or wrong way, just explore how you naturally would during a research process.
3. Please try to think out loud and talk throughout your walk-through with the prototype. Share any thoughts you may be having or expectations that may arise during your interactions.
4. We will be here to guide you, but will still leave room for you to interact with the prototype.
5. Now, with your permission we would like to record the session, please let us know if that is okay with you [Participant Answers]. Remember, can you opt out at any time during the usability test if need be.
6. Now, I will begin recording our session, and with the prototype opened please share the screen of only the prototype screen and begin! You can stop sharing your screen, once you have completed your navigation with the prototype.
7. You will start with the onboarding process of the prototype and continue to the service of our prototype.

## **Follow-up questions after usability test:**

1. Now that you have navigated through the prototype we would like you to share general thoughts, and feelings you had during your experience.
  - a. Answer
2. Did you have any moments of confusion upon navigating through the prototype?
  - a. Answer
3. Did you have any expectations upon navigating through the prototype?
  - a. Answer
4. What were your feelings surrounding the prototype's features?
  - a. Answer
5. How was the experience of utilising the prototype for research purposes?
  - a. Answer
6. Any last thoughts regarding the prototype?
  - a. Answer
7. This is the end of the usability test, thank you so much for your time during this process. We are very appreciative of your time, and all the feedback provided. Thank you!